

Day 54 (1) - Virtualization & Cloud

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Virtualization & Cloud Info

Although Cisco is more known for their networking devices, they also offer hardware servers such as Unified Computing System (UCS).

Servers before Virtualization

- Before virtualization, there was a one-to-one relationship between a physical server and an operating system.
- In that operating system, apps providing services such a web server, email server, etc. A physical server would be used for each of the services.
- This is inefficient for multiple reasons:
 - Physical servers are expensive and take up space, power, and etc.
 - The resources on each physical server are typically under-used.

Virtualization

Virtualization (Type 1 Hypervisor)

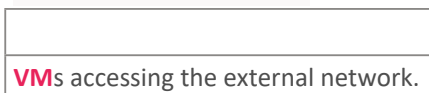
- Virtualization allows us to run multiple OS's on a single physical server.
- Each instance is called a **Virtual Machine (VM)**.
- A **hypervisor** is used to manage to allocate the hardware resources to each **VM**.
- Another name for a **hypervisor** is **Virtual Machine Monitor (VMM)**.
- **Type 1 hypervisor** is the type of **hypervisor** which runs directly on top of the hardware.
 - They're also called **bare-metal hypervisors**.
 - Another term is **native hypervisor**.

Virtualization (Type 2 Hypervisor)

- **Type 2 hypervisor** run as a program on an operating system (called a **Host OS**) like a regular computer program.
 - They're also called **hosted hypervisors**.
- The OS running directly on the hardware is called **Host OS**, and the OS running in a VM is called **Guest OS**.
- They are rarely used in data centers, but commonly used on personal-use devices.

Connecting VMs to the Network

- **VMs** are connected to each other and the external network via a virtual switch running on the **hypervisor**.
- Just like a regular physical switch, the **vSwitch** interfaces can operate as access or trunk ports and use VLANs to separate the **VMs** at **Layer 2**.
- Interfaces on the **vSwitch** connect to the physical **Network Interface Cards (NICs)** of the server to communicate with the external network.



Cloud Services

Traditional IT infrastructure deployments were some combination of the following:

1. On-Premises

- All devices are located on company property.
- All equipment is purchased and owned by the company.
- The company is the responsible for necessary space, power, cooling and etc.

2. Colocation

- Data centers that rent out space for customers to put their infrastructure.
- The equipment is still the responsibility of the end customer, although they are not located on the customer's premises.
- The data center is the one who provides the space, electricity, cooling and etc.

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Cloud Services Components

Cloud services are made up of many different components:

- **Five** essential characteristics
- **Three** service models
- **Four** deployment models

Five Essential Characteristics

1. **On-demand self-service**
 - The customer is able to use/stop the service freely (via a web portal) without direct communication to the service provider.
2. **Broad network access**
 - The service is available through standard network connections, and can be accessed through many kinds of devices.
3. **Resource pooling**
 - A pool of resources is provided by the service provider, and when a customer requests a service, the resources to fulfill that request are allocated from the shared pool.
4. **Rapid elasticity**
 - Customers can quickly expand the services they use in the cloud from a pool of resources that appears to be infinite. Likewise, they can quickly reduce their services when not needed.
5. **Measured service**
 - The cloud service provider measures the customer's usage of cloud resources, and the customer can measure their own use as well. Customers are charged based on usage.

Three Service Models

1. **Software as a Service (SaaS)**
 - Example(s):
 - Microsoft Office 365
2. **Platform as a Service (PaaS)**
 - Example(s):
 - AWS Lambda
 - Google App Engine
3. **Infrastructure as a Service (IaaS)**
 - Example(s):
 - Amazon EC2
 - Google Compute Engine

Four Deployment Models of Cloud

1. **Public Cloud**
 - The cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.
 - Popular **public cloud** providers include:
 - AWS
 - Microsoft Azure
 - GCP
 - OCI
 - IBM Cloud
 - Alibaba Cloud
2. **Private Cloud**
 - The cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.
3. **Community Cloud**
 - The cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

4. Hybrid Cloud

- The cloud infrastructure is a composition of two or more distinct cloud infrastructures (private, community, or public) that remain unique entities, but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load balancing between clouds).

Different ways of connecting to cloud resources